Pharmaceutical Calculations Booklet

Midterm · Lectures 1 – 7

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Each lecture is split into topics. Every topic has **Lecture Problems**, then **Textbook Problems**.

1 Introduction, Units & Equivalences

In this lecture

1. Proportional Sets & Ratios 4 lecture · 7 textbook
2. Fractions 1 lecture · 16 textbook
3. Decimals 1 lecture · 9 textbook
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9. Units, Prefixes & Equivalences 4 lecture · 15 textbook
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1.1 Proportional Sets & Ratios

Lecture Problems

Slide 3 A drug has a concentration of 100 mg/5 ml. Complete the table:

Weight of drug (mg)	Volume (ml)
20	?
50	?
75	?
150	?

a. Volume that contains 20 mg = ?

b. Volume that contains 50 mg = ?

c. Volume that contains 75 mg = ?

d. Volume that contains 150 mg = ?

Slide 3 A car travels at 100 km/hour. Complete the table:

Distance (km)	Time (hour)
50	?
150	?
200	?

a. Time for 50 km = ?

b. Time for 150 km = ?

c. Time for 200 km = ?

Slide 4 The upper set is proportional to the lower set:

9	21	30	36
3	7	10	12

What is the ratio of the upper set to the lower set?

Slide 4 The upper set is proportional to the lower set:

2	5	7
10	25	35

What is the ratio of the upper set to the lower set?

Textbook Problems

Examples 2.1–2.5 · Page 7 In each of the following, decide whether or not the pair of sets is proportional.

2.1

6	10	14
3	5	7

2.2

5	7	9
40	56	72

2.3

2	4	8
1	2	3

2.4

5	7	9
15	17	19

2.5

16	17	21
80	85	105

Q1 · Page 26 Decide whether or not the pair of sets is proportional.

6	8	10…
3	5	7…

Proportional or not?

Q2 · Page 26 Decide whether or not the pair of sets is proportional.

5	7	9…
20	28	36…

Proportional or not?

Q3 · Page 26 Decide whether or not the pair of sets is proportional.

2	3	4…
5	6	7…

Proportional or not?

Q4 · Page 26 Decide whether or not the pair of sets is proportional.

2	4	6…
5	10	15…

Proportional or not?

Q5 · Page 26 Decide whether or not the pair of sets is proportional.

25	35…
5	7…

Proportional or not?

Q6 · Page 26 Decide whether or not the pair of sets is proportional.

2	4	6…
3	6	9…

Proportional or not?

1.2 Fractions

Lecture Problems

Slide 5 x/3 = 16/24

Find what is x.

Textbook Problems

Example 2.6 · Page 9 Find the numerator of the fraction that is equal to 1/7 but has denominator 21.

Numerator = ?

Example 2.7 · Page 9 Find the numerator of the fraction that is equal to 16/24 but has denominator 3. (Same as the lecture question on Slide 5.)

Numerator = ?

Example 2.8 · Page 9 Find the denominator of the fraction that is equal to 80/180, but has numerator 16.

Denominator = ?

Q7 · Page 26 Find the fraction that is equal to 7/8 but has denominator 72.

Answer = ?

Q8 · Page 26 Find the fraction that is equal to 4/7 but has numerator 28.

Answer = ?

Q9 · Page 26 Find the fraction that is equal to 2/9 but has denominator 117.

Answer = ?

Example 2.9 · Page 10 Express 35/45 in its lowest terms.

Lowest terms = ?

Example 2.10 · Page 10 Express 60/405 in its lowest terms.

Lowest terms = ?

Q10 · Page 26 Express 12/20 in its lowest terms.

Answer = ?

Q11 · Page 26 Express 3/15 in its lowest terms.

Answer = ?

Q12 · Page 26 Express 42/54 in its lowest terms.

Answer = ?

Q13 · Page 26 Express 78/130 in its lowest terms.

Answer = ?

Q14 · Page 26 Express 120/42 in its lowest terms.

Answer = ?

Q15 · Page 26 Multiply 3/13 by 5.

Answer = ?

Q16 · Page 26 Multiply 3/17 by 4.

Answer = ?

Q17 · Page 27 Multiply 5/9 by 7.

Answer = ?

1.3 Decimals

Lecture Problems

Slide 6 The number 0.23 is a rational number: the 2 is in the tenths column and the 3 is in the hundredths column, so 0.23 = 23/100.

a. Convert 0.5 to a fraction.

---

b. Convert 0.44 to a fraction.

Textbook Problems

Example 2.11 · Page 11 Convert 0.3 to a fraction.

Fraction = ?

Example 2.12 · Page 12 Convert 0.35 to a fraction.

Fraction = ?

Q18 · Page 27 Convert 0.8 to a fraction in its lowest terms.

Answer = ?

Q19 · Page 27 Convert 0.87 to a fraction in its lowest terms.

Answer = ?

Q20 · Page 27 Convert 0.875 to a fraction in its lowest terms.

Answer = ?

Example 2.13 · Page 12 Convert the fraction 3/5 to a decimal.

Decimal = ?

Example 2.14 · Page 12 Convert the fraction 7/8 to a decimal.

Decimal = ?

Q21 · Page 27 Convert the fraction 5/8 to a decimal.

Answer = ?

Q22 · Page 27 Convert the fraction 7/80 to a decimal.

Answer = ?

1.4 Percentages

Lecture Problems

Slide 8 8 percent = 8 per 100 = 8/100

a. Convert 3/8 to percent.

b. Convert 0.125 to percent.

c. Convert 25% to fraction.

d. Convert 0.6% to fraction.

Textbook Problems

Example 2.19 · Page 14 Express 7/10 as a percentage.

Answer = ?

Example 2.20 · Page 14 Express 3/8 as a percentage.

Answer = ?

Q27 · Page 27 Express 7/8 as a percentage.

Answer = ?

Q28 · Page 27 Express 6/25 as a percentage.

Answer = ?

Example 2.15 · Page 13 Express 0.76 as a percentage.

Answer = ?

Example 2.16 · Page 13 Express 0.769 as a percentage.

Answer = ?

Q23 · Page 27 Express 0.82 as a percentage.

Answer = ?

Q24 · Page 27 Express 0.345 as a percentage.

Answer = ?

Example 2.17 · Page 13 Express 20% as a fraction in its lowest terms.

Answer = ?

Example 2.18 · Page 13 Express 0.2% as a fraction in its lowest terms.

Answer = ?

Q25 · Page 27 Express 30% as a fraction in its lowest terms.

Answer = ?

Q26 · Page 27 Express 0.3% as a fraction in its lowest terms.

Answer = ?

1.5 Finding a Missing Value & Practical Problems

Textbook Problems

Example 2.21 · Page 16 For the proportional sets:

7	y	9	10…
21	24	27	30…

Find the missing value y.

Example 2.22 · Page 17 For the proportional sets:

x	14
25	50

Find the missing number x.

Example 2.23 · Page 18 For the proportional sets:

3	x
5	8

Find the missing value x.

Example 2.24 · Page 19 For the pair of proportional sets:

z	9
6	15

Find the missing value z.

Example 2.25 · Page 19 For the proportional sets:

37	52
7	z

Find the missing number z.

Example 2.28 · Page 21 Find the value of p in the two proportional sets:

36	9
22	p

p = ?

Q29 · Page 27 Find the missing value x from the pair of proportional sets:

8	x	20
2	4	5

Missing value = ?

Q30 · Page 27 Find the missing number from the pair of proportional sets:

25	30
y	6

Missing value = ?

Q31 · Page 27 Find the missing number from the pair of proportional sets:

21	z
7	30

Missing value = ?

Q32 · Page 28 Find the missing number from the pair of proportional sets:

81	18
x	2

Missing value = ?

Q33 · Page 28 Find the missing number from the pair of proportional sets:

y	21
8	24

Missing value = ?

Q34 · Page 28 Find the missing number from the pair of proportional sets:

4	10
10	z

Missing value = ?

Q35 · Page 28 Find the missing number from the pair of proportional sets:

x	10
7	35

Missing value = ?

Q36 · Page 28 Find the missing number from the pair of proportional sets:

32	y
19	41

Give your answer correct to two significant figures.

Missing value = ?

Q37 · Page 28 Find the missing number from the pair of proportional sets:

225	420
192	z

Give your answer correct to three significant figures.

Missing value = ?

Q38 · Page 28 Find the missing number from the pair of proportional sets:

x	11.3
22.5	15.7

Give your answer correct to three significant figures.

Missing value = ?

Example 2.26 · Page 20 Consider three sets each proportional to each other:

x	20
y	50
30	150

Find x and y.

Example 2.27 · Page 21 Find the values of a and b in the following proportional sets:

a	25	110
12	b	22

a = ?, b = ?

Example 2.29 · Page 21 Find the values of x and y in the three proportional sets:

24	6
x	15
64	y

x = ?, y = ?

Example 2.30 · Page 22 Find the values of y and z in the two proportional sets:

y	18	21
9	z	63

y = ?, z = ?

Worked example · Page 23 A suspension contains 120 mg of paracetamol in each 5 mL. Find the volume of suspension that contains 300 mg of paracetamol.

Volume = ?

Example 2.31 · Page 23 The recommended dose of phenytoin for a patient is 840 mg. An injection contains 50 mg/mL. What volume of injection is needed for the recommended dose?

Volume = ?

Example 2.32 · Page 24 1 L of aqueous solution contains 250 mL of syrup. How much syrup will be contained in 25 L of the solution?

Volume of syrup = ?

Example 2.33 · Page 24 25 g of ointment contains 5 g of calamine. How much calamine will be contained in 60 g of ointment?

Mass of calamine = ?

1.6 Exponentials (Powers of 10)

Lecture Problems

Slide 9 Physical and chemical measurements deal with either very small or large numbers, so for convenience we use exponentials. Positive power means multiply; negative power means divide.

a. Write 3,000,000 using a power of 10.

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b. Write 0.000003 using a power of 10.

1.7 Ratio Strength, Percentage Strength & ppm

Lecture Problems

Slide 10 · Example 1 A percentage is the number of parts per one hundred and is probably the commonest expression of concentration used in pharmacy.

Express 1:2500 as a percentage strength.

Slide 11 · Example 2 Express 0.1% as a ratio strength.

Ratio strength = ?

Slide 11 · Example 3 Express 1 p.p.m. as a percentage strength.

Percentage strength = ?

1.8 Variation (Inverse Proportion)

Lecture Problems

Slide 12 Sometimes pharmaceutical calculations deal with inverse relationships — twice the cause, half the effect — as when you decrease the strength of a solution by increasing the amount of diluent.

If 10 litres of a 5% solution are diluted to 40 litres, what is the percentage strength of the dilution?

1.9 Units, Prefixes & Equivalences

Lecture Problems

Slide 14 Prefixes table — complete the **Factor** column:

Prefix	Symbol	Division	Factor
mega	M	× 1000,000	?
kilo	k	× 1000	?
deci	d	÷ 10	?
centi	c	÷ 100	?
milli	m	÷ 1000	?
micro	mc	÷ 1000,000	?
nano	n	÷ 1000,000,000	?

a. mega (M)

b. kilo (k)

c. deci (d)

d. centi (c)

e. milli (m)

f. micro (mc)

g. nano (n)

Slide 15 Complete the equivalences:

a. 1 L = ? ml

b. 1 kg = ? g = ? mg = ? mcg = ? ng

Slide 17 Examples — fill in the blanks:

a. 0.5 kg = ? g = ? mg

b. 0.25 g = ? mg (and in kg?)

c. 0.2 mg = ? mcg

d. 0.4 mg = ? kg

e. 0.5 L = ? ml

f. 0.25 mol = ? mmol

Slide 18 For drug calculation it is best to work in whole numbers (i.e. 125 mcg not 0.125 mg). To convert from a larger unit to a smaller unit: × 1000. From a smaller unit to a larger unit: ÷ 1000.

Write 0.125 mg as a whole number in a smaller unit.

Textbook Problems

Example 3.1 · Page 33 Change 200 mg to grams.

Answer = ?

Example 3.2 · Page 33 Convert 5473 mL to litres.

Answer = ?

Example 3.3 · Page 34 Convert 1.26 mg to micrograms.

Answer = ?

Example 3.4 · Page 34 Convert 135 465 micrograms to grams.

Answer = ?

Example 3.5 · Page 34 Convert 0.003 745 kL to mL.

Answer = ?

Q1 · Page 41 Change 0.035 g to milligrams.

Answer = ?

Q2 · Page 41 Change 1384 mg to grams.

Answer = ?

Q3 · Page 41 Change 437 mL to litres.

Answer = ?

Q4 · Page 42 Change 12.47 L to millilitres.

Answer = ?

Q5 · Page 42 Change 0.87 mL to microlitres.

Answer = ?

Q6 · Page 42 Change 15 750 microlitres to millilitres.

Answer = ?

Q7 · Page 42 Change 0.0025 kg to grams.

Answer = ?

Q8 · Page 42 Change 12 g to kilograms.

Answer = ?

Q9 · Page 42 Change 0.72 megalitres to kilolitres.

Answer = ?

Q10 · Page 42 Change 145 kg to megagrams.

Answer = ?

1.10 Conversions Between Systems & Temperature

Lecture Problems

Slide 19 (practice) Height: convert inch to cm. Weight: convert pound to kg. (The slide gives no numbers — these two use the standard factors 1 inch = 2.54 cm and 1 pound = 0.4536 kg.)

a. Convert 10 inches to cm.

---

b. Convert 150 pounds to kg.

Textbook Problems

Example 3.6 · Page 36 Convert 3.4 pints to litres. (1 L = 1.76 pints)

Litres = ?

Q11 · Page 42 Use the conversion factor 1 L = 1.76 pints to express:

(a) 6.43 pints in litres

(b) 2.42 L in pints

Example 3.7 · Page 37 Convert 106.4 kg to stones and pounds. (1 pound = 0.4536 kg; 14 pounds = 1 stone)

Stones and pounds = ?

Example 3.8 · Page 37 Convert 10 stone 4 pounds to kilograms. (1 pound = 0.4536 kg)

Kilograms = ?

Q12 · Page 42 Use the conversion factor 1 pound = 0.4536 kg to express:

(a) 9 stones 4 pounds in kilograms (1 stone = 14 pounds)

(b) 49 kg in stones and pounds

Example 3.9 · Page 38 Convert 5 feet 10 inches into metres. (1 inch = 25.4 mm = 0.0254 m)

Metres = ?

Example 3.10 · Page 38 What would be the BMI for a patient who weighs 10 stone and 4 pounds and is 5 feet 10 inches tall? (BMI = weight (kg) / height² (m))

BMI = ?

Worked example · Page 40 Find the number of degrees Celsius corresponding to 95 degrees Fahrenheit. (50 °C = 122 °F; water freezes at 0 °C = 32 °F)

°C = ?

Example 3.11 · Page 40 Change 25 degrees Celsius to degrees Fahrenheit. (0–100 °C corresponds to 32–212 °F, a range of 100 °C and 180 °F)

°F = ?

Example 3.12 · Page 41 Change 68 degrees Fahrenheit to degrees Celsius.

°C = ?

Q13 · Page 42 The boiling point of water is 100 degrees Celsius or 212 degrees Fahrenheit. The freezing point of water is 0 degrees Celsius or 32 degrees Fahrenheit. Express:

(a) 15 degrees Celsius in degrees Fahrenheit (to the nearest degree)

(b) 88 degrees Fahrenheit in degrees Celsius (to the nearest degree)

1.11 Dosage Calculations

Lecture Problems

Slide 21 The dose required is 3 mg/kg and the patient weighs 68 kg. This means that for every kg of patient's weight, you will need 3 mg of drug.

Total dose required = ?

Slide 22 In some cases, the dose is given as a total daily dose — it has to be given in divided doses (3 or 4 times a day).

Ibuprofen 1.2 g daily given 3 times daily. What is each single dose?

Slide 23 The dose for Cefuroxime iv is 2.25 g in divided doses every 8 hours.

What is each dose?

Slide 24 You need to prepare an infusion of digoxin containing 0.75 mg. Digoxin is available as 500 mcg/2 ml.

How many ml do you need?

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2 Drug Concentrations

In this lecture

1. Fractions, Decimals & Percentages 2 lecture
2. Calculations with % 3 lecture
3. Ratio Strength 1 lecture · 7 textbook
4. Percentage Concentration (w/v, w/w, v/v) 3 lecture · 16 textbook
5. mg/ml Concentrations & Amount Strength 2 lecture · 2 textbook
6. Parts per Million (ppm) 2 textbook
7. Converting Between Expressions of Concentration 6 textbook
8. Heparin & Insulin (Units) 1 lecture
9. Practice Calculations (Lecture) 3 lecture

2.1 Fractions, Decimals & Percentages

Lecture Problems

Slide 5 Fractions & percentages: to convert a fraction to a percentage, multiply by 100; to convert a percentage to a fraction, divide by 100.

a. Convert 2/8 to a percentage.

b. Convert 25% to a fraction.

Slide 6 Decimals & percentages: to convert a decimal to a percentage, multiply by 100; to convert a percentage to a decimal, divide by 100.

a. Convert 0.5 to a percentage.

b. Convert 50% to a decimal.

2.2 Calculations with %

Lecture Problems

Slide 7 How much is 35% in 500 ml?

Amount = ?

Slide 8 What is the percentage of 480 of 8000?

Percentage = ?

Slide 9 How many ml is 40% of 1.3 litre?

Volume (ml) = ?

2.3 Ratio Strength

Lecture Problems

Slide 10 Ratio strength is expressed as a ratio in the form of 1 in x.

5 g of product contains 250 mg of sulphur in yellow soft paraffin. Express this in ratio.

Textbook Problems

Example 4.2 · Page 49 2 L of an aqueous solution contains 50 mL of ethanol. Express this as a ratio strength.

Ratio strength = ?

Example 4.3 · Page 49 5 g of product contains 250 mg of sulfur in yellow soft paraffin. Express this as a ratio strength. (Same as the lecture question on Slide 10.)

Ratio strength = ?

Example 4.16 · Page 60 What is the amount of potassium permanganate in 300 mL of a 1 in 25 solution and what is the percentage strength of the solution?

Amount = ? and % = ?

Q4 · Page 61 How many milligrams of y are needed to make 200 mL of a 1 in 500 solution?

Answer = ?

Q5 · Page 61 How many millilitres of y are needed to produce 400 mL of a 1 in 200 solution?

Answer = ?

Q6 · Page 61 How many milligrams of y are needed to produce 25 g of a 1 in 5 ointment?

Answer = ?

Q7 · Page 61 How many grams of y are there in 250 mL of a 1 in 80 solution?

Answer = ?

2.4 Percentage Concentration (w/v, w/w, v/v)

Lecture Problems

Slide 11 The most common way of describing drug concentration is percentage concentration.

a. What does 3% w/v mean?

---

b. What does 3% w/w mean?

---

c. What does 3% v/v mean?

Slide 12 How much dextrose is there in a 500 ml infusion of dextrose 4% w/v?

Dextrose = ?

Slide 13 You have salbutamol nebuliser solution 0.3%.

What volume is required for a 4 mg dose?

Textbook Problems

Example 4.15 · Page 59 How many milligrams of aluminium acetate are required to prepare 500 mL of a 0.03% w/v solution?

Amount (mg) = ?

Q9 · Page 62 How many milligrams of x are needed to make 5 mL of an 8% solution?

Answer = ?

Q10 · Page 62 How many millilitres of x are needed to make 600 mL of a 3% solution?

Answer = ?

Q11 · Page 62 How many milligrams of x are needed to make 200 mL of a 3.4% solution?

Answer = ?

Q12 · Page 62 How many milligrams of x are there in 100 mL of a 0.01% solution?

Answer = ?

Q13 · Page 62 How much calamine is required to produce 250 g of a 3% ointment?

Answer = ?

Q8 · Page 62 Potassium permanganate 0.1 g; Purified water to 100 mL.

How many millilitres of a 2.5% potassium permanganate solution could be used in place of the 0.1 g in the above formula?

Q14 · Page 62 What volume of 17% w/v solution contains 1.5 g of ingredient?

Answer = ?

Q15 · Page 62 What volume of 20% w/v solution contains 5 g of ingredient?

Answer = ?

Q16 · Page 62 What volume of 15% w/v solution contains 7 g of ingredient?

Answer = ?

Q17 · Page 62 What volume of 34% w/v solution contains 150 g of ingredient?

Answer = ?

Example 4.6 · Page 52 A cream contains 12 g of drug X made up to 100 g with cream base. What is the percentage concentration?

Percentage = ?

Example 4.8 · Page 53 Express 900 mg of sodium chloride made up to 100 mL with water as a percentage.

Percentage = ?

Q3 · Page 61 If the following amounts of drug are made up to 5 g with lactose what is the percentage concentration of the resulting mix?

(a) 100 mg

(b) 150 mg

(c) 200 mg

(d) 300 mg

(e) 500 mg

Example 4.7 · Page 52 Express 1 in 500 w/v solution of potassium permanganate as a percentage.

Percentage = ?

Example 4.9 · Page 53 A morphine sulfate injection contains 10 mg/mL. What is the percentage concentration?

Percentage = ?

2.5 mg/ml Concentrations & Amount Strength

Lecture Problems

Slide 14 Sometimes it may be useful to convert percentage concentrations to mg/ml.

Express sodium chloride 0.9% in mg/ml.

Slide 16 You can find the strength in mg/ml by multiplying the percentage by 10; to convert mg/ml to percentage, divide by 10.

a. Convert 0.9% to mg/ml.

b. Convert 9 mg/ml to a percentage.

Textbook Problems

Example 4.1 · Page 47 A preparation contains 900 mg of sodium chloride dissolved in water to produce 100 mL of solution. Express the concentration of the solution as an amount strength (mg/mL and g/mL).

Amount strength = ?

Example 4.12 · Page 56 A solution contains a solid dissolved in a liquid. The ratio strength is 1 in 2000 w/v. What are the percentage strength and the amount concentration expressed as mg/mL?

% = ? and mg/mL = ?

2.6 Parts per Million (ppm)

Textbook Problems

Example 4.4 · Page 50 Fluoride supplements should not be taken if the amount of fluoride in the water supply exceeds 0.7 parts per million w/v. Express this ratio in mg/L.

mg/L = ?

Example 4.5 · Page 51 If the concentration of fluoride is 0.25 ppm w/v, how many litres would contain 1 mg of fluoride?

Litres = ?

2.7 Converting Between Expressions of Concentration

Textbook Problems

Example 4.10 · Page 54 A solution contains 20 mL of ethanol in 500 mL of product. Express the concentration as a ratio strength and as a percentage strength.

Ratio = ? and % = ?

Example 4.11 · Page 55 A solid ingredient mixed with a solid vehicle has a ratio strength of 1 in 40. Find the percentage strength and the amount strength expressed as grams per gram.

% = ? and g/g = ?

Q1 · Page 60 Convert the following ratio strengths into percentage strengths:

(a) 1 in 25

(b) 1 in 20

(c) 1 in 50

(d) 1 in 800

(e) 1 in 500

(f) 1 in 2000

(g) 1 in 300

Example 4.13 · Page 57 A liquid ingredient mixed with another liquid vehicle has a concentration of 5% v/v. Find the ratio strength and the amount strength expressed as mL/mL.

Ratio = ? and mL/mL = ?

Example 4.14 · Page 57 5 g of solid ingredient is added to 45 g of a base. Find the percentage strength, the ratio strength and the amount strength expressed as g/g. (Same as the lecture question on Slide 22.)

% = ?, ratio = ?, g/g = ?

Q2 · Page 61 What is the concentration of the solutions, expressed as percentage strength and ratio strength, when the following amounts of drug Z are dissolved in enough water to produce 125 mL of solution?

(a) 25 g

(b) 50 g

(c) 60 g

(d) 5 g

(e) 7 g

2.8 Heparin & Insulin (Units)

Lecture Problems

Slide 18 Heparin is available as 1000, 5000, 10000 and 25000 units/ml (1 ml ampules).

You need to give heparin as continuous infusion 13000 units in 35 ml normal saline over 24 hours. What do you withdraw?

2.9 Practice Calculations (Lecture)

Lecture Problems

Slide 20 Practice Calculation 1

How many grams of sodium chloride are there in one litre infusion of sodium chloride 0.9%?

Slide 21 Practice Calculation 2

You need to give infusion of calcium gluconate 3 g. You have 10 ml ampules of calcium gluconate 10%. How much do you need to withdraw?

Slide 22 Practice Calculation 3

5 g of solid ingredient is added to 45 g of a base. Express as percentage, ratio, and as amount of strength g/g.

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3 Dilutions

In this lecture

1. Simple Dilutions 2 lecture · 2 textbook
2. Stock Solutions & Serial Dilutions 3 lecture · 9 textbook
3. Concentrated Waters 1 lecture · 1 textbook
4. Multiple Dilutions 1 lecture · 12 textbook
5. Triturations 1 lecture · 1 textbook
6. Powder Calculations 1 lecture · 3 textbook
7. Mixing Concentrations 2 lecture · 7 textbook

3.1 Simple Dilutions

Lecture Problems

Slides 4-5 You need to prepare 500 ml of 45% solution. What volume of stock solution do you need which when diluted to 500 ml will give 45% solution?

Volume of stock = ? and volume of water = ?

Slides 6-7 *Student activity:* 300 ml of 1 in 25 w/v solution diluted to 800 ml. What is the concentration of the diluted product in % and mg/ml?

a. % w/v = ?

---

b. mg/ml = ?

---

c. Ratio strength = ?

Textbook Problems

Example 5.2 · Page 67 100 mL of a 1 in 50 w/v solution is diluted to 1000 mL. Find the concentration of the diluted product as a percentage strength, a ratio strength and an amount strength expressed as mg/mL.

% , ratio and mg/mL = ?

Example 5.1 · Page 66 A product consists of a solid ingredient in a solid vehicle; 2 g of the solid ingredient is contained in 500 g of product. A further 250 g of vehicle is added to the product. Find the concentration of the diluted mixture as a percentage strength, a ratio strength and an amount strength in mg/g.

% , ratio and mg/g = ?

3.2 Stock Solutions & Serial Dilutions

Lecture Problems

Slides 8-9 You need to prepare 1 litre of 20% solution from 80% stock solution.

Stock volume = ? and diluent = ?

Slide 10 You need to prepare one litre of potassium chloride solution 1 in 10,000. You have a stock solution 10 g/L. How much of the stock solution do you need?

Stock volume = ?

Slide 13 To what volume must 350 ml of 40% w/v solution be diluted to produce a 12% solution?

Final volume = ?

Textbook Problems

Example 5.4 · Page 69 What volume of a 40% v/v solution needs to be used to produce 500 mL of 5% v/v solution?

Volume = ?

Q10 · Page 93 How many millilitres of 90% alcohol when diluted to 135 mL produces 60% alcohol? (Assume no contraction in volume.)

Volume = ?

Example 5.3 · Page 68 What volume of a 1 in 400 v/v solution is needed to produce 5 L of a 1 in 2000 v/v solution?

Volume = ?

Q12 · Page 94 What volume of a 0.6% w/v solution is required to produce 200 mL of 1 in 10 000 solution?

Volume = ?

Q13 · Page 94 What volume of a 1 in 50 solution is required to produce 300 mL of a 0.3% w/v solution?

Volume = ?

Example 5.5 · Page 70 To what volume must 250 mL of a 25% w/v solution be diluted to produce a 10% solution?

Final volume = ?

Q6 · Page 93 How many millilitres of water must be added to 180 mL of 40% v/v solution in order to produce a 5% v/v solution?

Water = ?

Q9 · Page 93 What volume of 100 mg/5 mL solution is required to be diluted to give 200 mL of 75 mg/5 mL solution?

Volume = ?

Q16 · Page 94 Calculate the volume of 0.2% w/v potassium permanganate solution required to produce 1000 mL of a 100 ppm solution.

Volume = ?

3.3 Concentrated Waters

Lecture Problems

Slide 11 Concentrated waters (rose water, peppermint water, chloroform water) are used to produce single-strength products. They are intended for dilution in a ratio of 1 in 40.

What does "1 in 40" mean here?

Textbook Problems

Worked example · Page 72 Make volumes of single-strength chloroform water (50 mL, 100 mL, 200 mL, 300 mL, 500 mL) from chloroform water concentrate (1 : 40).

a. Concentrate (mL) for each volume = ?

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b. Double-strength chloroform water — concentrate for the same volumes = ?

3.4 Multiple Dilutions

Lecture Problems

Slide 12 *Student activity:* What weight of ingredient is required to produce 500 ml of a solution, when 3 ml diluted to 40 ml, produces a strength of 0.5% w/v solution?

Weight = ?

Textbook Problems

Example 5.10 · Page 81 What weight of ingredient is required to produce 1000 mL of a solution such that, when 2.5 mL of it is diluted to 50 mL with water, it gives a 0.25% w/v solution?

Weight = ?

Example 5.12 · Page 83 What weight of ingredient is required to produce 250 mL of a solution such that 10 mL of this solution diluted to 150 mL gives a 0.1% w/v solution?

Weight = ?

Q1 · Page 93 What weight of potassium permanganate is required to produce 300 mL of solution such that 5 mL of this solution diluted to 250 mL gives a 0.01% w/v solution?

Weight = ?

Q2 · Page 93 What weight of potassium permanganate is required to produce 500 mL of solution such that 25 mL of this solution diluted to 500 mL gives a 0.025% w/v solution?

Weight = ?

Q3 · Page 93 What weight of zinc sulfate is required to produce 150 mL of solution such that 10 mL of this solution diluted to 200 mL gives a 1% w/v solution?

Weight = ?

Q8 · Page 93 What weight of potassium permanganate is needed to produce 300 mL of solution such that if 1 mL is diluted to 20 mL a 0.1% w/v solution is produced?

Weight = ?

Example 5.11 · Page 82 What weight of malachite green oxalate is required to produce 500 mL of solution such that 25 mL of this solution diluted to 4000 L gives a 1 in 2 000 000 solution?

Weight = ?

Q4 · Page 93 What weight of potassium permanganate is required to produce 200 mL of solution such that 50 mL of this solution diluted to 200 mL will produce a 1 in 400 solution?

Weight = ?

Q7 · Page 93 What weight of copper sulfate is required to produce 150 mL of solution such that 30 mL of this solution diluted to 150 mL produces a 1 in 6000 solution?

Weight = ?

Example 5.13 · Page 84 How many millilitres of 0.5% w/v solution are required to produce 250 mL of 1 in 5000 solution?

Volume = ?

Example 5.14 · Page 85 How many millilitres of a 1 in 80 w/v solution are required to make 500 mL of a 0.02% solution?

Volume = ?

Q5 · Page 93 What weight of malachite green oxalate is required to produce 300 mL of a solution such that 25 mL of this solution diluted to 2000 L gives a 0.5 ppm solution?

Weight = ?

3.5 Triturations

Lecture Problems

Slide 14 You need to prepare 100 ml of the following formula:

Hyoscine hydrobromide	500 mcg
Water to	5 ml

a. Amount of hyoscine in 100 ml = ?

---

b. Minimum weight on the scale is 100 mg. How do you prepare it?

Textbook Problems

Example 5.6 · Page 73 How would you prepare 100 mL of a preparation to the following formula?

hyoscine hydrobromide (micrograms)	500
chloroform water (mL) to	5

(Same as the lecture question on Slide 14.)

How do you prepare it?

3.6 Powder Calculations

Lecture Problems

Slide 15 Powders must weigh a minimum of 120 mg. Prepare seven powders each containing 80 mg of drug x.

Drug, diluent and total weight for 1 and 7 powders = ?

Textbook Problems

Example 5.7 · Page 76 Prepare five powders each containing 100 mg of paracetamol.

Weights = ?

Example 5.8 · Page 76 Prepare five powders each containing 10 mg of hyoscine hydrobromide.

How do you prepare them?

Example 5.9 · Page 78 Prepare six powders each containing 0.5 mg of drug.

How do you prepare them?

3.7 Mixing Concentrations

Lecture Problems

Slide 16 What is the final strength (% v/v) if 200 ml of 40% v/v solution is added to 300 ml of 70% v/v solution?

Final % v/v = ?

Slide 17 What proportion of 60% v/v solution and 10% v/v solution would produce a 500 ml of 40% v/v solution mixture?

V1 and V2 = ?

Textbook Problems

Example 5.15 · Page 86 What is the final % v/v of a solution if 200 mL of 40% v/v solution is added to 300 mL of 70% v/v solution? (Same as the lecture question on Slide 16.)

Final % v/v = ?

Q11 · Page 94 When 70 mL of sodium chloride 0.9% w/v solution is combined with 100 mL of 1.7% w/v sodium chloride solution what is the final strength of the solution?

Final strength = ?

Q14 · Page 94 Calculate the volumes of two stock solutions of 100 mg/5 mL and 25 mg/5 mL that must be mixed to produce 500 mL of 75 mg/5 mL solution.

Volumes = ?

Q15 · Page 94 What volumes of two stock solutions of 30% w/v and 60% w/v are required to make 300 mL of 40% w/v solution?

Volumes = ?

Example 5.16 · Page 88 A suspension used to be provided as two strengths, both 100 mg/5 mL and 25 mg/5 mL solutions, which could be mixed to produce intermediate doses. If we want to produce a product that has a strength of 75 mg/5 mL, what proportion of the two mixtures do we need?

Proportion = ?

Example 5.17 · Page 90 Find the proportions of the same two suspensions given in Example 5.16 that would be required to produce a 50 mg/5 mL suspension.

Proportion = ?

Example 5.18 · Page 91 What proportions of 90% v/v and 50% v/v ethanol mixtures would produce a 70% v/v mixture? Assume no contraction.

Proportion = ?

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4 Moles

In this lecture

1. Moles & Molecular Mass 3 lecture · 2 textbook
2. Amount of an Ion in a Compound 1 lecture · 8 textbook
3. Drug Salts, Bases & Hydrates 1 lecture · 15 textbook
4. Legal Classification Using Molecular Weights 3 textbook
5. Moles & Millimoles 1 lecture · 5 textbook
6. Millimoles in Infusions & Molar Solutions 4 lecture · 7 textbook

4.1 Moles & Molecular Mass

Lecture Problems

Slides 3-4 One mole = the relative atomic mass in grams. One mole of any element contains 6×10^{23} atoms (Avogadro's constant). Atomic masses: Ca = 40, Cl = 35.5, Na = 23, C = 12, H = 1.

a. How many moles are in 24 g of carbon?

b. How many moles (and atoms) are in 1 g of carbon?

Slide 5 Sodium chloride (NaCl) is made up of sodium (Na) and chloride (Cl): $NaCl \rightarrow Na^{+} + Cl^{-}$.

Molecular mass of NaCl = ? and what does 1 mole of NaCl give?

Slide 6 CO_2: C = 12, O = 16.

Molecular mass of CO_2 = ?

Textbook Problems

Worked examples · Page 166 Calculate the molecular weights (K = 39, Br = 79.9, Al = 27, Cl = 35.5, Zn = 65.4, S = 32, O = 16, H = 1, C = 12):

a. Potassium bromide, KBr

---

b. Aluminium chloride, $AlCl_3$

---

c. Zinc sulfate monohydrate, $ZnSO_4.H_2O$

---

d. Aspirin, $C_9H_8O_4$

Q1 · Page 189 For the following drugs calculate the molecular weight and the weight of salt containing 1 millimole of the metal ion:

a. (a) potassium chloride, KCl

b. (b) sodium chloride, NaCl

c. (c) calcium chloride, $CaCl_2.2H_2O$

d. (d) calcium lactate, $C_6H_{10}CaO_6.5H_2O$

e. (e) potassium bicarbonate, $KHCO_3$

f. (f) magnesium chloride, $MgCl_2.6H_2O$

g. (g) sodium citrate, $C_6H_5Na_3O_7.2H_2O$

h. (h) ammonium chloride, NH_4Cl

4.2 Amount of an Ion in a Compound

Lecture Problems

Slide 13 How many mg of sodium ions are contained in 850 mg tablet of sodium chloride?

Na^+ (mg) = ?

Textbook Problems

Example 10.1 · Page 167 How many milligrams of sodium ions are contained in a 600-mg tablet of sodium chloride?

Na^+ (mg) = ?

Q8 · Page 190 Calcium acetate, $C_4H_6CaO_4$.

a. Molecular weight = ?

---

b. Calcium provided by 300 mg of calcium acetate = ?

---

c. mg of calcium acetate in 1 millimole = ?

Q11 · Page 190 Sodium Fluoride BP drops contain 275 micrograms of sodium fluoride per drop. Calculate the quantity of fluoride ion provided by five drops (Na = 23, F = 19).

Fluoride = ?

Example 10.2 · Page 168 What is the percentage of lithium contained in a 400-mg tablet of lithium carbonate?

% Li = ?

Q2 · Page 189 Lithium citrate, $C_6H_5Li_3O_7.H_2O$ (Li = 7).

a. Molecular weight = ?

---

b. % of Li^+ = ?

Q3 · Page 190 The empirical formula of ferrous fumarate is $C_4H_2FeO_4$.

a. Molecular weight = ?

b. Weight of ferrous fumarate that gives 100 mg of iron = ?

Q4 · Page 190 A syrup contains ferrous fumarate 140 mg/5 mL. How much iron will a patient receive per day if the daily dosage regimen is 10 mL twice a day?

Iron per day = ?

Q7 · Page 190 Calcium gluconate (molecular weight 448.4) tablets each contain 53.4 mg of calcium (Ca = 40.08).

a. Calcium gluconate in each tablet = ?

b. Millimoles of calcium per tablet = ?

4.3 Drug Salts, Bases & Hydrates

Lecture Problems

Slide 14 What weight of ferrous gluconate contains the same quantity of iron as tablet containing 300 mg of ferrous sulphate (dried)? ($FeSO_4$ = 152, ferrous gluconate = 482, Fe = 56)

Ferrous gluconate = ?

Textbook Problems

Example 10.3 · Page 169 What weight of ferrous gluconate contains the same quantity of iron as a tablet containing 200 mg of ferrous sulfate (dried)?

Ferrous gluconate = ?

Example 10.4 · Page 171 Erythromycin 734, erythromycin ethyl succinate 862, erythromycin stearate 1018. Calculate the amounts of erythromycin ethyl succinate and erythromycin stearate that are equivalent to 250 mg erythromycin.

Ethyl succinate and stearate = ?

Example 10.5 · Page 172 Adrenaline injection contains 100 micrograms of adrenaline (as the acid tartrate) per mL. MW adrenaline 183, adrenaline acid tartrate 333. Calculate the amount of adrenaline acid tartrate in each mL.

Acid tartrate per mL = ?

Q12 · Page 191 Ranitidine Injection BP contains ranitidine 25 mg/mL (as the hydrochloride). Calculate the amount of ranitidine hydrochloride needed to prepare 100 mL. (MW: ranitidine hydrochloride 350.9, ranitidine 314.4)

Ranitidine HCl = ?

Q17 · Page 191 A patient requires 300 mg of quinine (MW 324.4). What weight of quinine hydrobromide (MW 423.3) will provide 300 mg of quinine?

Quinine HBr = ?

Example 10.6 · Page 173 How many grams of salbutamol sulfate will contain the same amount of salbutamol base as 5 g of salbutamol? MW salbutamol 239, salbutamol sulfate $(C_{13}H_{21}NO_3)_2.H_2SO_4$ 577.

Salbutamol sulfate = ?

Example 10.7 · Page 174 BNF: quinine (anhydrous base) 100 mg ≡ quinine bisulfate 169 mg ≡ quinine dihydrochloride 122 mg. Quinine base MW = 324. Estimate the molecular weights of the two salts.

MW of bisulfate and dihydrochloride = ?

Example 10.8 · Page 174 Show that a 300-mg tablet of quinine bisulfate contains less quinine than a 300-mg tablet of quinine sulfate (BNF equivalent for quinine sulfate = 121).

Quinine base in each = ?

Example 10.9 · Page 176 1 L of Paediatric Ferrous Sulfate Oral Solution BP contains 12 g of ferrous sulfate. You need 150 mL, but only ferrous sulfate anhydrous is available. MW ferrous sulfate 278, anhydrous 152. How much anhydrous should be used?

Anhydrous ferrous sulfate = ?

Example 10.10 · Page 177 How much codeine phosphate sesquihydrate (MW 424) is equivalent to 30 mg of codeine phosphate (hemihydrate, MW 406)?

Sesquihydrate = ?

Q6 · Page 190 Glucose intravenous infusion contains 5% glucose expressed as anhydrous glucose.

a. Anhydrous glucose (MW 180) for 500 mL = ?

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b. Glucose monohydrate (MW 192) for 750 mL = ?

Q14 · Page 191 A syrup contains chloroquine sulfate (MW 436), 68 mg/5-mL dose. Calculate the amount of chloroquine base (MW 319.9) in each 5-mL dose.

Base = ?

Q15 · Page 191 How much chloroquine base (MW 319.9) is equivalent to a tablet containing 250 mg chloroquine phosphate (MW 515.9)?

Base = ?

Q16 · Page 191 How much morphine (MW 285.3) is contained in 50 mL of a 2% solution of morphine tartrate (MW 774.8)?

Morphine = ?

Q18 · Page 191 How much hyoscine (MW 303.4) is contained in a 20-mg tablet of hyoscine butylbromide (MW 440.4)?

Hyoscine = ?

4.4 Legal Classification Using Molecular Weights

Textbook Problems

Example 10.11 · Page 178 Ephedrine HCl can be sold over the counter if the maximum dose ≡ 30 mg ephedrine and the maximum daily dose ≡ 60 mg ephedrine. MW ephedrine HCl 202, ephedrine 165. What are these limits as ephedrine hydrochloride?

Max dose and max daily dose = ?

Example 10.12 · Page 179 Morphine salts are CD Inv POM up to 0.2% (as anhydrous morphine base). MW base 285, morphine HCl 376. Maximum morphine hydrochloride in 200 mL?

Morphine HCl = ?

Example 10.13 · Page 180 Codeine is CD Inv POM up to 2.5% (as base) or 100 mg base per dosage unit. MW codeine base 317, codeine phosphate 406.

a. Max codeine phosphate in 100 mL syrup = ?

---

b. Max codeine phosphate per dosage unit = ?

4.5 Moles & Millimoles

Lecture Problems

Slide 7 In practice, moles are too big for everyday use, so millimoles are used.

1 millimole = ? mole

Textbook Problems

Example 10.14 · Page 182 How many millimoles each of Na^+ and HCO_3^- are there in a 500-mg sodium bicarbonate capsule? ($NaHCO_3$ = 84)

mmol = ?

Example 10.15 · Page 183 Calculate the number of moles of chloride ion in 500 mg of calcium chloride ($CaCl_2$ = 110).

Chloride = ?

Q9 · Page 190 Calcium phosphate, $Ca_3(PO_4)_2$.

a. Molecular weight = ?

---

b. A powder contains 3.3 g. Millimoles of calcium ion = ?

Example 10.16 · Page 184 Calculate the number of moles of water in 500 mg of $CaCl_2.6H_2O$ (MW 219).

Water = ?

Q13 · Page 191 "Low sodium" means less than 1 millimole of sodium per tablet or 10-mL dose. What is the maximum amount of sodium chloride in a 10-mL dose?

NaCl = ?

4.6 Millimoles in Infusions & Molar Solutions

Lecture Problems

Slides 8-9 How many millimoles of sodium are there in 500 ml infusion containing 1.8 mg/ml sodium chloride?

mmol Na = ?

Slide 10 How many millimoles of sodium are there in a 1 litre infusion containing 1800 mg/litre sodium chloride?

mmol = ?

Slide 11 How many millimoles of sodium are in one litre sodium chloride infusion 0.9%?

mmol = ?

Slide 12 *Student activity:* You need to add 35 mmol of potassium chloride to one litre of infusion. You have ampule containing potassium chloride 2 g/10 ml. Calculate how much do you need to withdraw? (KCl = 39 + 35.5 = 74.5)

Volume = ?

Textbook Problems

Example 10.19 · Page 186 How many millimoles of sodium and chloride ions are contained in 1 L of Sodium Chloride Infusion BP (0.9%)? (Same as the lecture question on Slide 11.)

mmol = ?

Q5 · Page 190 A syrup contains 7.5% of potassium chloride. How many millimoles of K^+ are contained in 1 mL?

mmol K^+ = ?

Q10 · Page 190 Magnesium sulfate injection is 50% magnesium sulfate ($MgSO_4.7H_2O$). How many millimoles of magnesium ion per mL?

mmol/mL = ?

Example 10.17 · Page 185 How many moles of solute are there in 20 mL of a 1 mol/L solution?

Moles = ?

Example 10.18 · Page 185 How much sodium chloride is required to make 25 mL of a 0.5 molar solution? How many moles of sodium ion will there be?

NaCl and Na^+ = ?

Example 10.21 · Page 188 How much sodium fluoride (MW 42) is required to make 250 mL of a mouthwash containing 0.012 mol/L?

NaF = ?

Example 10.20 · Page 187 How many millimoles/L of chloride ions are in Potassium Chloride (0.3%) and Sodium Chloride (0.9%) Infusion BP? (KCl 74.5, NaCl 58.5)

Chloride = ?

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5 Doses for Adults

In this lecture

1. Solid Dose Units (Tablets, Capsules...) 1 lecture · 22 textbook
2. Oral Liquid Formulations 1 lecture · 13 textbook
3. Dose Volumes Less Than 5 ml 1 lecture · 4 textbook
4. Injections & Units 6 textbook
5. Doses Based on Body Weight 3 lecture
6. Doses Based on Body Surface Area (BSA) 3 lecture

5.1 Solid Dose Units (Tablets, Capsules...)

Lecture Problems

Slide 8 A prescription requires two capsules of drug X for 5/7. How many tablet should be supplied?

Number = ?

Textbook Problems

Example 7.1 · Page 113 Calculate the total number of tablets to be dispensed if a prescription requires two tablets of drug X three times a day for 7 days.

Tablets = ?

Example 7.2 · Page 114 A prescription requires two suppositories of drug Y daily for 3/7. How many suppositories should be supplied?

Suppositories = ?

Q1 · Page 125 A patient is prescribed two tablets of drug A four times a day for 3 months. How many tablets?

Tablets = ?

Q7 · Page 125 Charcoal sachets: two sachets three times a day for 1/12. How many sachets?

Sachets = ?

Example 7.6 · Page 117 A prescription requires a patient to receive 20 mg domperidone (in tablet form) every 8 hours for 4 weeks. Domperidone is available only as 10-mg tablets. How many tablets?

Tablets = ?

Q23 · Page 127 A scored tablet contains 100 mg of drug X. Prescription: 50 mg three times a day for 10 days. How many tablets?

Tablets = ?

Q24 · Page 127 Drug Z is only available as a 62.5-mg capsule. Prescription: 125 mg four times a day for 1/12. How many capsules?

Capsules = ?

Q25 · Page 127 Drug A is available as a 500-microgram tablet. A patient requires 2 mg four times a day for 3/7 and 1.5 mg for 5/7.

a. Tablets for a 2-mg dose = ?

---

b. Tablets for a 1.5-mg dose = ?

---

c. Total tablets = ?

Q26 · Page 127 Drug B: 500 mg twice a day for 5 days; available as 250-mg capsules. How many capsules?

Capsules = ?

Example 7.7 · Page 118 A patient requires 8 mg of drug X per day for 1 week followed by 7 mg per day for the next week. Drug X is available in tablets of 1 mg and 4 mg. How many tablets of each strength?

Tablets = ?

Q27 · Page 127 Drug A is available as 2.5-mg and 5-mg tablets. Prescription: 7.5 mg TDS for 5 days, then 5 mg QDS for 3 days, then 5 mg BD for 3 days, then 2.5 mg TDS for 7 days. How many of each?

Tablets = ?

Q2 · Page 125 A customer buys a pack of 100 vitamin capsules. The dosage regimen is one capsule three times a day. How many days will the pack last?

Days = ?

Q3 · Page 125 An inhaler contains 200 metered doses. If the patient inhales two doses three times a day, how long will the inhaler last?

Days = ?

Q6 · Page 125 A residential home buys a 1000 pack of senna tablets. Each of 35 residents takes two tablets each night. How long will it last?

Days = ?

Q4 · Page 125 Analgesic suppositories to be used six times a day for 5 days, then three times a day for 3 days. How many suppositories?

Suppositories = ?

Q8 · Page 125 Drug Y: three capsules twice a day for 3/7, two capsules twice a day for 3/7, one capsule twice a day for 2/52. How many capsules?

Capsules = ?

Q5 · Page 125 A patient needs 8 weeks' antimalarial treatment. Drug A: one tablet per week; drug B: two tablets daily. How many tablets of A and B?

Tablets = ?

Q11 · Page 126 One nicotine patch every 3 days. How many patches for 3 months?

Patches = ?

Q9 · Page 126 A sachet of 0.9% saline is used to bathe an eye. Bathe both eyes twice a day for 1/12. How many sachets?

Sachets = ?

Q10 · Page 126 A patient going on holiday for 3 weeks is supplied 168 tablets. Regimen: two tablets four times a day. Enough?

Enough?

Q12 · Page 126 A pack contains 56 tablets. Regimen: one tablet four times a day for 4 weeks. How many packs?

Packs = ?

Q13 · Page 126 A pack of 84 tablets is designed to last 1 month. If the patient takes one dose twice a day, what is the dose?

Dose = ?

5.2 Oral Liquid Formulations

Lecture Problems

Slide 9 A prescription of indigestion mixture with dosage regimen of 20 ml four times daily for 5 days.

A. How many 5 ml spoonful of mixture required for each dose?

B. Calculate the number of doses.

C. Calculate the total volume of mixture that should be supplied.

Textbook Problems

Example 7.3 · Page 115 Indigestion mixture: 15 mL four times a day for 5 days. How many 5-mL spoonfuls per dose? Number of doses and total volume?

Spoonfuls, doses, volume = ?

Example 7.4 · Page 115 A 100-mL bottle of antihistamine mixture; dosage two 5-mL spoonfuls four times a day. How many days will it last?

Days = ?

Q29 · Page 128 Drug X: 240 mg four times a day; send 100 mL of 120 mg/5 mL mixture. How many days will it last?

Days = ?

Example 7.8 · Page 119 360 mg of paracetamol four times a day for 3 days, supplied as 120 mg/5 mL suspension. Volume to supply?

Volume = ?

Q28 · Page 127 Drug Z: 62.5 mg three times a day for 5 days; available as 250 mg/5 mL suspension. Volume of each dose and total?

Dose volume and total = ?

Q31 · Page 128 Elixir 150 mg/5 mL. Prescription: 225 mg four times a day for 7 days. Volume to supply?

Volume = ?

Q14 · Page 126 How many 5-mL doses are contained in 120 mL of mixture?

Doses = ?

Q17 · Page 126 How many 15-mL doses are contained in 240 mL of mixture?

Doses = ?

Q15 · Page 126 30 mL of mouthwash twice a day for a 14-day course. Volume to supply?

Volume = ?

Q16 · Page 126 Indigestion mixture: 15 mL four times a day for 2/52. How much to dispense?

Volume = ?

Q18 · Page 126 10 mL of laxative solution at night. Volume used in a month?

Volume = ?

Q21 · Page 127 7.5 mL of a mixture four times a day for 3/52. Volume to supply?

Volume = ?

Q19 · Page 126 A 150-mL bottle of cough elixir. Each of two children gets 5 mL four times a day for 3 days. How many doses left?

Doses left = ?

5.3 Dose Volumes Less Than 5 ml

Lecture Problems

Slide 12 A prescription states: 2.5 ml three times daily for 4/7. What volume of drug should be dispensed?

Volume = ?

Textbook Problems

Q20 · Page 126 A prescription for a young child reads: 2.5 mL three times a day for 4/7. What volume should be dispensed? (Same as the lecture question on Slide 12.)

Volume = ?

Example 7.5 · Page 117 A very young child is prescribed 2.0 mL of an analgesic suspension three times a day for 4/7. How many mL to dispense?

Volume = ?

Example 7.9 · Page 120 What volume of paracetamol suspension (250 mg/5 mL) would provide a dose of 200 mg?

Volume = ?

Q22 · Page 127 A pack contains 30 mL of a syrup. How many days will it last if the dose is 0.5 mL three times a day?

Days = ?

5.4 Injections & Units

Textbook Problems

Example 7.10 · Page 120 A patient requires an IM injection of 0.5 mg of adrenaline. Adrenaline injection contains 100 micrograms/mL. How many mL?

Volume = ?

Q32 · Page 128 15 mg of morphine sulfate SC every 4 hours for 16 hours; injection 20 mg/mL. Volume per dose and total morphine?

Volume per dose and total = ?

Example 7.11 · Page 122 Nystatin: 500 000 units every 6 hours. Suspension 100 000 units/mL. Total daily dose and volume for 5 days?

Daily dose and volume = ?

Q30 · Page 128 A solution contains 500 units/mL; pack size 25 mL. 2500 units three times a day for 5 days. How many packs?

Packs = ?

Example 7.12 · Page 123 Vitamin A drops contain 5000 units/mL; 40 drops/mL. How many drops give 750 units?

Drops = ?

Example 7.13 · Page 124 Product A: 8 mg vitamin A (retinyl acetate) + 400 IU ergocalciferol. Product B: 2500 IU vitamin A + 10 micrograms vitamin D. Which contains the most vitamins? (Vit A: 1 IU = 0.344 micrograms; Vit D: 40 IU = 1 microgram)

Which product?

5.5 Doses Based on Body Weight

Lecture Problems

Slide 15 The usual dose of chlorambucil is 150 mcg/kg once a day. How many milligrams should be administered to a person weighing 154 lb?

Dose (mg) = ?

Slide 16 Normal gentamicin dose is 3 mg/kg/day in three divided doses. What would be the single (8-hour) dose for a patient weighing 165 lb scheduled to receive only 40% of the usual dose (renal impairment)?

Single dose = ?

Slide 17 A patient weighing 120 lb was administered 2.1 g of a drug supposed to be dosed at 30 mg/kg. Correct, overdose or underdose?

Verdict?

5.6 Doses Based on Body Surface Area (BSA)

Lecture Problems

Slides 18-20 BSA = √(Ht (cm) × Wt (kg) / 3600) (average adult = 1.73 m^2). If the dose of a drug is 5 mg/m^2, what would be the dose for a patient measuring 165 cm and weighing 65 kg?

Dose = ?

Slide 21 The chemotherapy regimen "CAF" during a 21-day cycle is: cyclophosphamide 500 mg/m^2, D-1; doxorubicin 50 mg/m^2, D-1; 5-fluorouracil 500 mg/m^2, D-1, 8. Calculate the dose of each drug per cycle for a patient with a BSA of 1.9 m^2.

Doses per cycle = ?

Slide 22 VAD regimen (multiple myeloma), 28-day cycle: vincristine 0.4 mg/d CIVI, D 1–4; doxorubicin 9 mg/m²/d CIVI, D 1–4; dexamethasone 40 mg/d PO, D 1–4, 9–12, 17–20. Total of each drug per cycle for BSA 1.65 m²?

Totals = ?

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6 Doses for Children

In this lecture

1. Rules Based on Age & Weight 2 lecture · 1 textbook
2. Doses Based on Body Weight 3 lecture · 9 textbook
3. Doses Based on Body Surface Area (BSA) 3 lecture · 5 textbook
4. Other Dispensing Calculations 4 lecture

6.1 Rules Based on Age & Weight

Lecture Problems

Slide 4 Drug dosage based on age — write the rule:

a. Young's rule

b. Cowling's rule

c. Fried's rule for infants

Slide 5 Drug dosage based on weight — write Clark's rule.

Clark's rule

Textbook Problems

Worked example · Page 132 BNF: "for most drugs the adult maximum dose should not be exceeded". A drug has a dose of 8 mg/kg and a maximum dose of 300 mg.

Dose for a 10-kg child and a 40-kg child = ?

6.2 Doses Based on Body Weight

Lecture Problems

Slide 9 Rx Phenytoin 30 mg/5 ml Suspension, 4 mg/kg/day for a 25-lb child.

Daily dose (mg) and volume (ml) = ?

Slide 14 The usual dose of sulfisoxazole for infants over 2 months and children is 60 to 75 mg/kg. What would be the usual range for a child weighing 44 lb?

Range = ?

Slide 16 A pediatric patient receives enalaprilat IV every 12 hours: 5 mcg/kg → 55 mcg per dose. Convert to oral enalapril 100 mcg/kg as a single daily dose. A 2.5-mg tablet is crushed and made up to 12.5 mL with sterile water. Dose in mL?

Volume = ?

Textbook Problems

Q2 · Page 138 Pethidine HCl injection 100 mg/2 mL. Child IM dose 0.5–2 mg/kg. Both extremes for a 12-year-old and the volumes.

Range and volumes = ?

Example 8.1 · Page 133 Fluconazole for mucosal candidiasis in children is 3 mg/kg daily. Calculate the dose for a child aged 3 years and a child aged 11 years and suggest a formulation (capsules 50, 150, 200 mg; suspension 50 mg/5 mL, 200 mg/5 mL).

a. 3-year-old

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b. 11-year-old

Q3 · Page 138 Griseofulvin oral dose for a child is 10 mg/kg daily; suspension 125 mg/5 mL. Dose and volume for a 5-year-old.

Dose and volume = ?

Q4 · Page 139 Using Q3, calculate the oral dose of griseofulvin and the volume of suspension for a 12-year-old child.

Dose and volume = ?

Example 8.2 · Page 135 Co-trimoxazole IV infusion: 36 mg/kg daily in two divided doses. Dose for a 5-year-old child; sterile solution 96 mg/mL.

Dose and volume = ?

Example 8.3 · Page 136 Amikacin: 15 mg/kg daily in two divided doses. Dose for a 6-month-old child; paediatric injection 50 mg/mL.

Dose and volume = ?

Q6 · Page 139 Drug X: oral dose for a 6-month-old of 8 mg/kg twice a day; suspension 40 mg/5 mL. Dose and volume each time?

Dose and volume = ?

Q1 · Page 138 Morphine sulfate for a 6-month-old child by SC injection: 200 micrograms/kg. Quantity for one dose and volume of 10 mg/mL injection?

Dose and volume = ?

Q9 · Page 139 Hydroxychloroquine sulfate oral dose for a child: 6.5 mg/kg daily. Dose for a 7-year-old?

Dose = ?

6.3 Doses Based on Body Surface Area (BSA)

Lecture Problems

Slide 6 Drug dosage on body surface area: BSA = √(Ht (cm) × Wt (kg) / 3600) (or use nomogram).

a. Child dose from the adult dose = ?

b. Child dose if the dose per m² is given = ?

Slide 8 The usual pediatric dose of ephedrine sulfate is 25 mg/m². Calculate the dose for a child weighing 18 kg and measuring 82 cm in height.

Dose = ?

Slide 15 If the usual adult dose of a drug is 100 mg, determine the approximate dose for a child with a BSA of 0.83 m².

Dose = ?

Textbook Problems

Example 8.4 · Page 137 Oral dose of methotrexate for a 5-year-old child who weighs 18 kg. Oral dose 15 mg/m² weekly.

Weekly dose = ?

Example 8.5 · Page 138 Calculate the dose of drug X for a 3-year-old boy if the dose is 120 mg/m² each day.

Dose = ?

Q7 · Page 139 The dose of drug X is 75 mg/m². Dose for a 5-year-old child of average weight?

Dose = ?

Q8 · Page 139 Using Q7, calculate the dose of drug X for a 10-year-old child of average weight.

Dose = ?

Q5 · Page 139 Didanosine 120 mg/m² every 12 hours. Dose and daily dose for a 7-year-old child who weighs 25 kg.

Dose and daily dose = ?

6.4 Other Dispensing Calculations

Lecture Problems

Slide 10 Rx Theophylline 80 mg/15 ml, give 1.5 mg as directed.

Volume = ?

Slide 11 How many drops would be prescribed in each dose of a liquid medicine if 15 mL contained 60 doses? The dispensing dropper calibrates 32 drops/ml.

Drops = ?

Slide 12 How many milliliters of a liquid medicine would provide a patient with 2 table spoonfuls twice a day for 8 days?

Volume = ?

Slide 13 It takes approximately 2 g of ointment to cover a child's leg. Ointment for total leg eczema, applied twice a day for 1 week. Which size: 15 g, 30 g, 60 g, 120 g?

Size = ?

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7 Creatinine Clearance

In this lecture

1. Cockroft-Gault & Choosing the Weight 2 lecture
2. Ideal Body Weight (IBW) 3 lecture
3. Adjusted Body Weight (AjBW) 1 lecture
4. Creatinine Clearance & Dosing 2 lecture
5. Adjusting CrCl for Body Surface Area 1 lecture

7.1 Cockroft-Gault & Choosing the Weight

Lecture Problems

Slide 3 Cockroft-Gault equation — write it and the units.

CrCl (mL/min) = ?

Slide 4 Which weight is used in the Cockroft-Gault equation?

Weight by BMI = ?

7.2 Ideal Body Weight (IBW)

Lecture Problems

Slide 5 The ideal body weight (IBW) — write the formulas.

a. Males

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b. Females

Slide 6 What is the IBW of a man who is 174 cm tall and weighs 70 kilograms?

IBW = ?

Slide 7 What is the IBW for a woman who is 64 inches tall and weighs 70 kilograms?

IBW = ?

7.3 Adjusted Body Weight (AjBW)

Lecture Problems

Slides 8–9 AjBW (obese patients, actual weight > IBW) = IBW + 0.4 × (actual body weight − IBW). Calculate the AjBW of a woman who is 157 cm tall and weighs 80 kilograms.

AjBW = ?

7.4 Creatinine Clearance & Dosing

Lecture Problems

Slide 10 Determine the loading and maintenance doses of gentamicin for a 76-year-old male weighing 190 lb with a height of 6 feet and serum creatinine of 2.4 mg/dL (212 mcmol/L). Loading dose 1.5 mg/kg of IBW; maintenance 1 mg/kg of IBW every 8 hours after the initial dose.

a. Loading and maintenance doses = ?

b. Extra: his creatinine clearance (Cockroft-Gault) = ?

Slide 11 What is the creatinine clearance for a 40-year-old male patient with S.Cr of 4 mg/dL (354 µmol/L)? The patient is 5 ft 4 in in height and weighs 103 kg.

CrCl = ?

7.5 Adjusting CrCl for Body Surface Area

Lecture Problems

Slide 12 Adjusted CrCl = BSA / 1.73 × eCrCl. If a patient weighing 120 lb and measuring 60 in. in height has a calculated creatinine clearance of 40 mL per minute, adjust the CrCl based on body surface area.

Adjusted CrCl = ?